

26TH ANNIVERSARY OF
PCI-IOSS
SPINE SURGERY SUMMIT
in conjunction with
ISASS AP 2024

INTERNATIONAL SOCIETY FOR THE ADVANCEMENT OF SPINE SURGERY ASIA-PACIFIC SECTION

28 - 31 AUGUST 2024 | YOGYAKARTA, INDONESIA

Objective Functional Evaluations Following Endoscopic Spine Surgery: The Prospective Use of **Smart Wearable Technology**

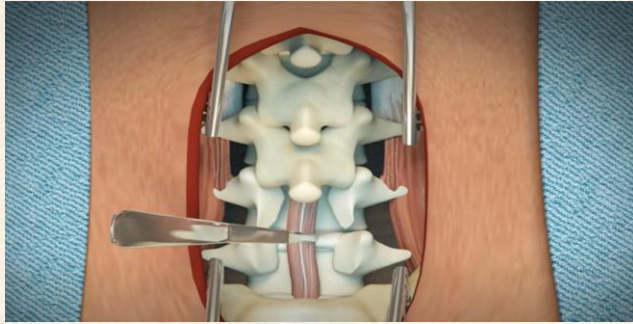
- A. A. Ngurah Nata Baskara, Yudha Mathan Sakti,
- Zikrina A. Lanodiyu, Tedjo Rukmoyo

Department of Orthopaedics and Traumatology
Sardjito General Hospital, Yogyakarta, Indonesia

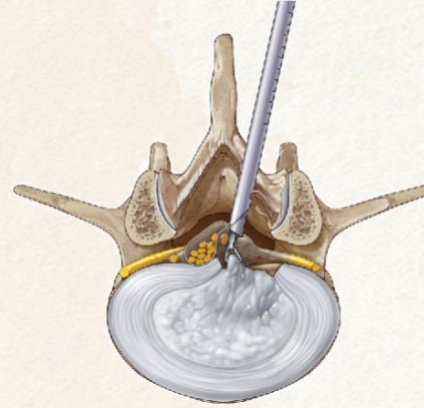


Background

Surgical treatment of the lumbar disc herniation has evolved

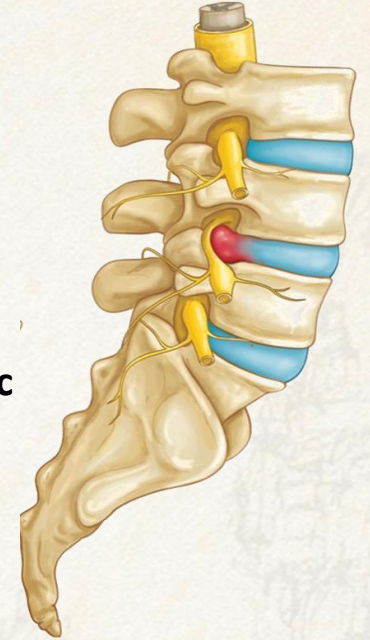


open surgery



PELD

"Percutaneous Endoscopic Lumbar Discectomy"

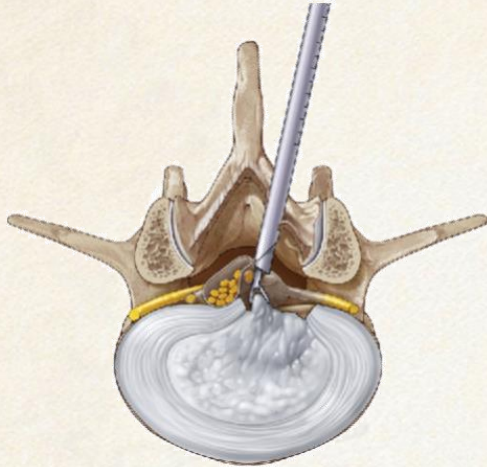


- **Minimal trauma and blood loss**
- **Shorter operation time**
- **Short hospital stay**
- **Early return to work** after surgery
- **Minimal damage to bony elements**
- **Preserving the stabilization** of the spine
- **Reduced risk of secondary degenerative diseases**

advantages



Background



PELD

Early mobilization after surgery is associated with improved survival, decreased length of hospitalization, and improved psychological well-being.

Available Parameters evaluating post PELD improvement :

Oswestry Disability Index (ODI)

SECTION 1 - PAIN INTENSITY

☐ I have no pain at the moment.
☐ The pain is very mild at the moment.
☐ The pain is moderate at the moment.
☐ The pain is fairly severe at the moment.
☐ The pain is very severe at the moment.
☐ The pain is the worst imaginable at the moment.

SECTION 2 - PERSONAL CARE (washing, dressing etc.)

☐ I can look after myself normally, without causing extra pain.
☐ I can look after myself normally, but it is very painful.
☐ It is painful to look after myself and I am slow and careful.
☐ I need some help, but manage most of my personal care.
☐ I need help every day in most aspects of self-care.
☐ I do not get dressed, wash with difficulty and stay in bed.

SECTION 3 - LIFTING

☐ I can lift heavy weights without extra pain.
☐ I can lift heavy weights, but it gives extra pain.
☐ Pain prevents me from lifting heavy weights off the floor, but I can manage if they are conveniently positioned (e.g., on a table).
☐ Pain prevents me from lifting heavy weights but I can manage light to medium weights if they are conveniently positioned.
☐ I can lift only very light weights.
☐ I cannot lift or carry anything at all.

SECTION 4 - WALKING

☐ Pain does not prevent me walking any distance.
☐ Pain prevents me walking more than 1 mile.
☐ Pain prevents me walking more than 1/2 of mile.
☐ Pain prevents me walking more than 100 yards.
☐ I can only walk using a stick or crutches.
☐ I am in bed most of the time and have to crawl to the toilet.

SECTION 5 - SITTING

☐ I can sit in any chair as long as I like.
☐ I can sit in my favourite chair as long as I like.
☐ Pain prevents me from sitting for more than 1 hour.
☐ Pain prevents me from sitting more than 1/2 an hour.
☐ Pain prevents me from sitting more than 10 minutes.
☐ Pain prevents me from sitting at all.

SECTION 6 - STANDING

☐ I can stand as long as I want without extra pain.
☐ I can stand as long as I want but it gives me extra pain.
☐ Pain prevents me from standing for more than 1 hour.
☐ Pain prevents me from standing for more than 1/2 an hour.
☐ Pain prevents me from standing for more than 10 minutes.
☐ Pain prevents me from standing at all.

SECTION 7 - SLEEPING

☐ My sleep is never disturbed by pain.
☐ My sleep is occasionally disturbed by pain.
☐ Because of pain, I have less than 6 hours of sleep.
☐ Because of pain, I have less than 4 hours of sleep.
☐ Because of pain, I have less than 2 hours of sleep.
☐ Pain prevents me from sleeping at all.

SECTION 8 - SEX LIFE (if applicable)

☐ My sex life is normal and causes no extra pain.
☐ My sex life is normal but causes some extra pain.
☐ My sex life is nearly normal but is very painful.
☐ My sex life is severely restricted by pain.
☐ My sex life is nearly absent because of pain.
☐ Pain prevents any sex life at all.

SECTION 9 - SOCIAL LIFE

☐ My social life is normal and causes me no extra pain.
☐ My social life is normal, but increases the degree of pain.
☐ Pain has no significant effect on my social life apart from limiting my more energetic interests, e.g., sports, etc.
☐ Pain has restricted my social life and I do not go out as often.
☐ Pain has restricted my social life to my home.
☐ I have no social life because of pain.

SECTION 10 - TRAVELLING

☐ I can travel anywhere without pain.
☐ I can travel anywhere but it gives extra pain.
☐ Pain is bad but I manage journeys over 2 hours.
☐ Pain restricts me to journeys of less than 1 hour.
☐ Pain restricts me to short necessary journeys under 30 minutes.
☐ Pain prevents me from travelling except to receive treatment.

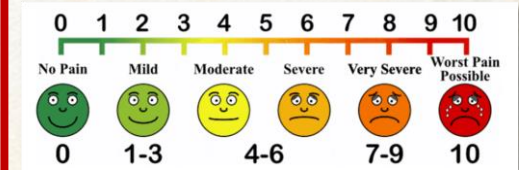
measures how back pain affected ability in daily life

Sciatica Bothersomeness Index (SBI)

	Not bothersome		Somewhat bothersome				Extremely bothersome
	0	1	2	3	4	5	6
Leg pain (sciatica)	☐	☐	☐	☐	☐	☐	☐
Numbness or tingling in leg, foot, or groin	☐	☐	☐	☐	☐	☐	☐
Weakness in leg or foot (e.g., difficulty lifting foot)	☐	☐	☐	☐	☐	☐	☐
Back or leg pain while sitting	☐	☐	☐	☐	☐	☐	☐

measures bothersomeness of Sciatic pain

Visual Analog Scale (VAS)



measures level of pain



Background

however...

**Questionnaire
Based**

Subjective

Language Barrier

**Recall and
Reporting bias**

**One time
assessment**

Early mobiliza
length of hos

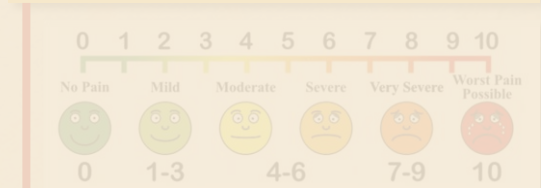
ated with improved survival, decreased
psychological well-being.

Oswestry Disab

evaluating post PELD success :

persomeness

**Visual Analog Scale
(VAS)**



measures level of pain

measures how
affected ability in daily life

measures bothersomeness

Background

however...

Questionnaire
Based

Subjective

Language Barrier

Recall and
Reporting bias

One time
assessment

[Eur Spine J.](#) 2011 Oct; 20(10): 1669–1675. Published online 2011 Apr 24. doi: [10.1007/s00586-011-1809-3](#)

PMCID: PMC3175874 | PMID: [21516463](#)

Estimates of success in patients with sciatica due to lumbar disc herniation depend upon outcome measure

[Anne Julsrud Haugen](#),^{M1} [Lars Grøvre](#),¹ [Jens Ivar Brox](#),² [Bård Natvig](#),^{3,7} [Anne Keller](#),⁴ [Dag Soldal](#),⁵ and [Margreth Grotle](#)^{6,3}

► [Author information](#) ► [Article notes](#) ► [Copyright and License information](#) [PMC Disclaimer](#)

The MDC for the ODI and RMDQ was, respectively, 13.67 and 4.87; the ES was 0.53 and 0.68; and the SRM was 0.80 and 0.81. ROC analysis revealed an area under the curve of 0.71 for the ODI and 0.64 for the RMDQ, thus indicating discriminating capacity; the best cut-off point for the dichotomous outcome was 9.5 for the ODI (sensitivity 76% and specificity 63%) and 2.5 for the RMDQ (sensitivity 62% and specificity 55%). These estimates were comparable between the subacute and chronic subjects. Both the ODI and RMDQ were significantly correlated with the SF-36 physical functioning subscale.

Highly subjective, with
varying sensitivity and
specificity

[Eur Spine J.](#) 2012 Jan; 21(1): 122–129. doi: [10.1007/s00586-011-1959-3](#)

PMCID: PMC3252446 | PMID: [21823035](#)

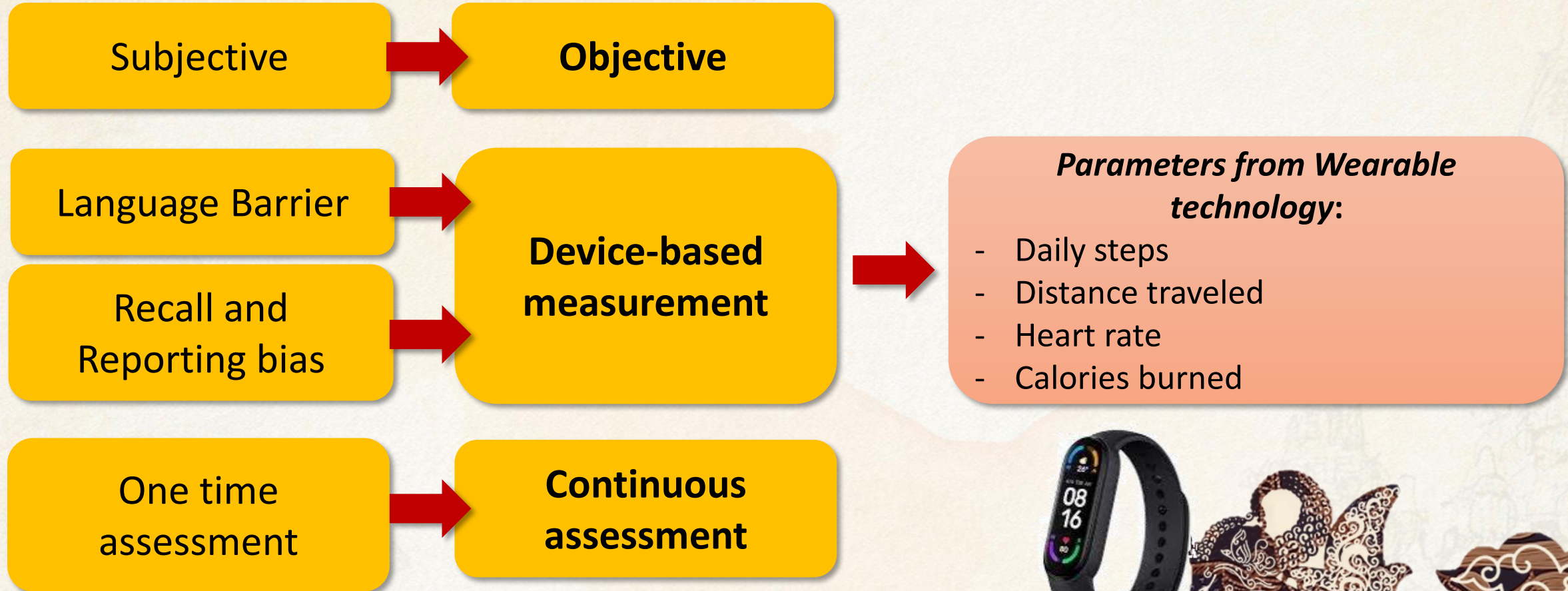
Responsiveness of the Oswestry Disability Index and the Roland Morris Disability Questionnaire in Italian subjects with sub-acute and chronic low back pain

[Marco Monticone](#),^{M1} [Paola Baiardi](#),² [Carla Vanti](#),³ [Silvano Ferrari](#),⁴ [Paolo Pillastrini](#),⁵ [Raffaele Mugnai](#),⁵ and [Calogero Foti](#)⁶

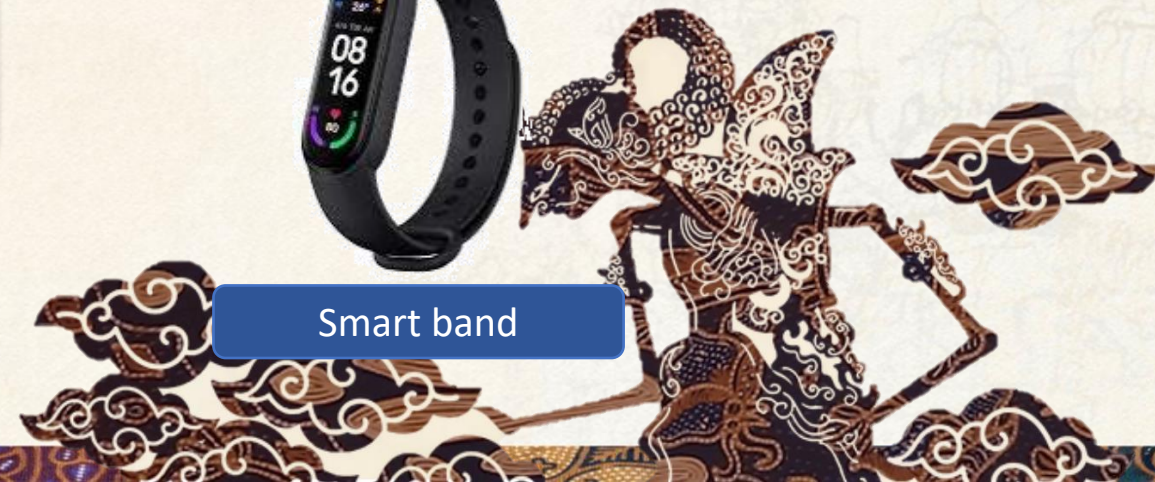
► [Author information](#) ► [Article notes](#) ► [Copyright and License information](#) [PMC Disclaimer](#)

Sciatica Bothersomeness Index 6.5 (0–24), Maine-Seattle Back Questionnaire 4.5 (0–12), and the SF-36 subscales bodily pain 51.5, and physical functioning 81.7 (0–100, higher values indicate better health). In conclusion, the success rates at 12 months varied from 49 to 58% depending on the measure used.

Modern solution :



Smart band



Modern solution :

 OPEN ACCESS PEER-REVIEWED CHAPTER

Wearable Technology for Assessment and Surgical Assistance in Minimally Invasive Surgery

Submitted: 08 September 2021 , Reviewed: 24 September 2021 , Published: 18 October 2021

DOI: 10.5772/intechopen.100617

WRITTEN BY

Juan A. Sánchez-Margallo, José Castillo Rabazo, Carlos Plaza de Miguel, Peter Gloor, David Durán Rey, Manuel Ramón González-Portillo, Isabel López Agudelo and Francisco M. Sánchez-Margallo

 Digital Medicine

www.nature.com/npjdigitalmed

ARTICLE **OPEN**

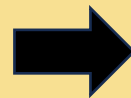
 Check for updates

Mobile devices and wearable technology for measuring patient outcomes after surgery: a systematic review

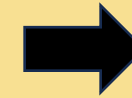
Stephen R. Knight¹ , Nathan Ng², Athanasios Tsanas³, Kenneth Mclean¹, Claudia Pagliari³  and Ewen M. Harrison¹

Objective factors

Advancement in
technology



New tools may aid in
evaluation following surgery



Use **in conjunction** with
available subjective tools

increase sensitivity and specificity in
predicting outcomes following PELD



Research Concept

SECTION 1 - PAIN INTENSITY

☐ I have no pain at the moment.
☐ The pain is very mild at the moment.
☐ The pain is moderate at the moment.
☐ The pain is fairly severe at the moment.
☐ The pain is very severe at the moment.
☐ The pain is the worst imaginable at the moment.

SECTION 2 - PERSONAL CARE (washing, dressing etc.)

☐ I can look after myself normally, without causing extra pain.
☐ I can look after myself normally, but it is very painful.
☐ It is painful to look after myself and I am slow and careful.
☐ I need some help, but manage most of my personal care.
☐ I need help every day in most aspects of self-care.
☐ I do not get dressed, wash with difficulty and stay in bed.

SECTION 3 - LIFTING

☐ I can lift heavy weights without extra pain.
☐ I can lift heavy weights, but it gives extra pain.
☐ Pain prevents me from lifting heavy weights off the floor, but I can manage if they are conveniently positioned (e.g. on a table).
☐ Pain prevents me from lifting heavy weights but I can manage 'light to medium' weights (they are conveniently positioned).
☐ I can lift only very light weights.
☐ I cannot lift or carry anything at all.

SECTION 4 - WALKING

☐ Pain does not prevent me walking any distance.
☐ Pain prevents me walking more than 1 mile.
☐ Pain prevents me walking more than 1/2 mile.
☐ Pain prevents me walking more than 100 yards.
☐ I can only walk using a stick or crutches.
☐ I am in bed most of the time and have to crawl to the toilet.

SECTION 5 - SITTING

☐ I can sit in any chair as long as I like.
☐ I can sit in my favourite chair as long as I like.
☐ Pain prevents me from sitting for more than 1 hour.
☐ Pain prevents me from sitting more than 1/2 an hour.
☐ Pain prevents me from sitting more than 10 minutes.
☐ Pain prevents me from sitting at all.

SECTION 6 - STANDING

☐ I can stand as long as I want without extra pain.
☐ I can stand as long as I want but it gives me extra pain.
☐ Pain prevents me from standing for more than 1 hour.
☐ Pain prevents me from standing for more than 1/2 an hour.
☐ Pain prevents me from standing for more than 10 minutes.
☐ Pain prevents me from standing at all.

SECTION 7 - SLEEPING

☐ My sleep is never disturbed by pain.
☐ My sleep is occasionally disturbed by pain.
☐ Because of pain, I have less than 6 hours of sleep.
☐ Because of pain, I have less than 4 hours of sleep.
☐ Because of pain, I have less than 2 hours of sleep.
☐ Pain prevents me from sleeping at all.

SECTION 8 - SEX LIFE (if applicable)

☐ My sex life is normal and causes no extra pain.
☐ My sex life is normal but causes some extra pain.
☐ My sex life is nearly normal but is very painful.
☐ My sex life is severely restricted by pain.
☐ My sex life is nearly absent because of pain.
☐ Pain prevents any sex life at all.

SECTION 9 - SOCIAL LIFE

☐ My social life is normal and causes no extra pain.
☐ My social life is normal, but increases the degree of pain.
☐ Pain has no significant effect on my social life apart from limiting my more energetic interests, e.g. sports, etc.
☐ Pain has restricted my social life and I do not go out as often.
☐ Pain has restricted my social life to my home.
☐ I have no social life because of pain.

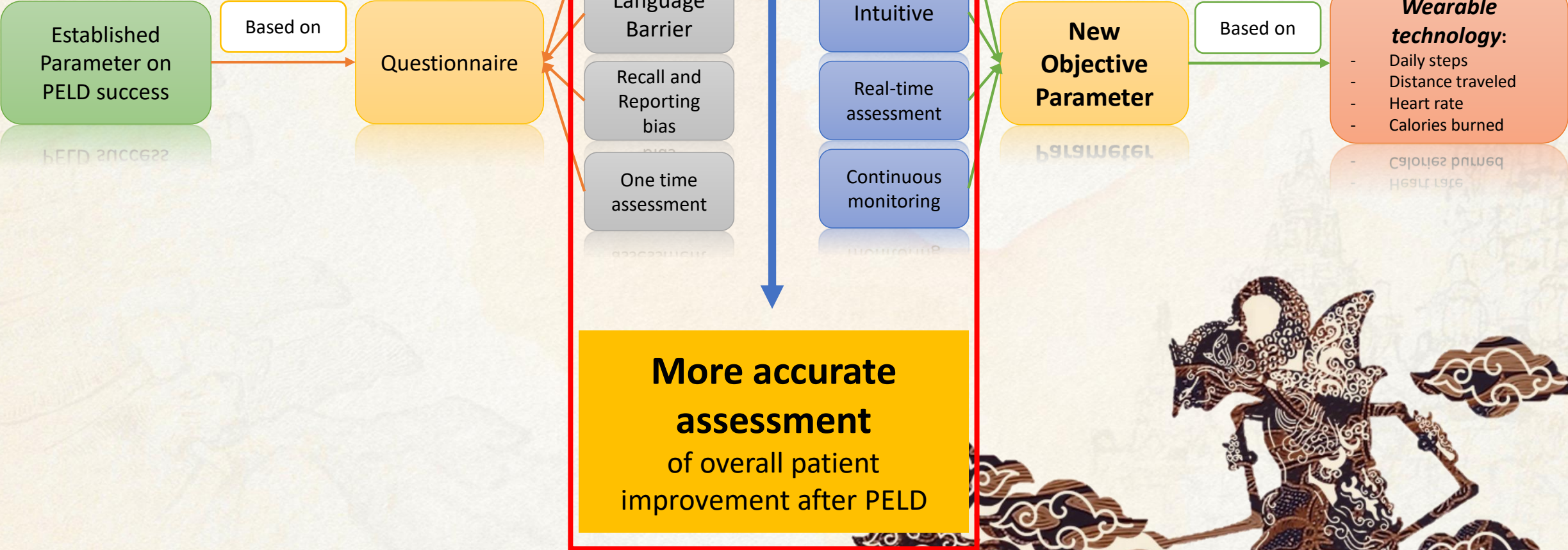
SECTION 10 - TRAVELLING

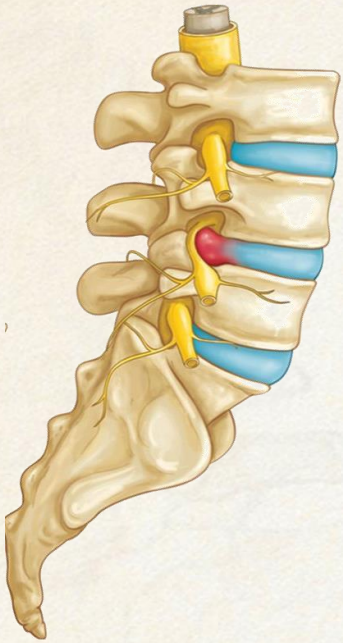
☐ I can travel anywhere without pain.
☐ I can travel anywhere but it gives extra pain.
☐ Pain is bad but I manage journeys over 2 hours.
☐ Pain restricts me to journeys of less than 1 hour.
☐ Pain restricts me to short necessary journeys under 30 minutes.
☐ Pain prevents me from travelling except to receive treatment.

SECTION 11 - LEG PAIN

☐ Leg pain (sciatica)
☐ Numbness or tingling in leg, foot, or groin
☐ Weakness in leg or foot (e.g. difficulty lifting foot)
☐ Back or leg pain while sitting

	Not bothersome		Somewhat bothersome		Extremely bothersome	
	0	1	2	3	4	5
Leg pain (sciatica)	☐	☐	☐	☐	☐	☐
Numbness or tingling in leg, foot, or groin	☐	☐	☐	☐	☐	☐
Weakness in leg or foot (e.g. difficulty lifting foot)	☐	☐	☐	☐	☐	☐
Back or leg pain while sitting	☐	☐	☐	☐	☐	☐





Research Questions

Is there a correlation between **steps, distance traveled and average heart rate changes** with **VAS, ODI score, and SBI score** in patient with lumbar disc herniation following the **Interlaminar PELD**?



Objectives

To determine the correlation between **steps, distance traveled and average heart rate changes** with **VAS, ODI score, and SBI score** in patient with lumbar disc herniation following the **Interlaminar PELD**

To produce a **predictive scoring system** based on the **objective parameter** to determine patient's improvement

Materials and Methods

Prospective cohort study



Conducted in Sardjito General Hospital



Between **July 2023 – December 2023**

Sampling method : **consecutive sampling**

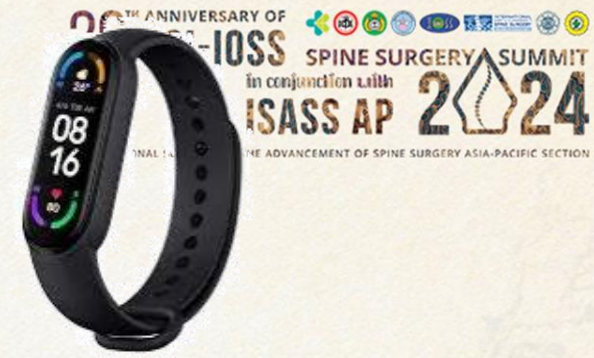
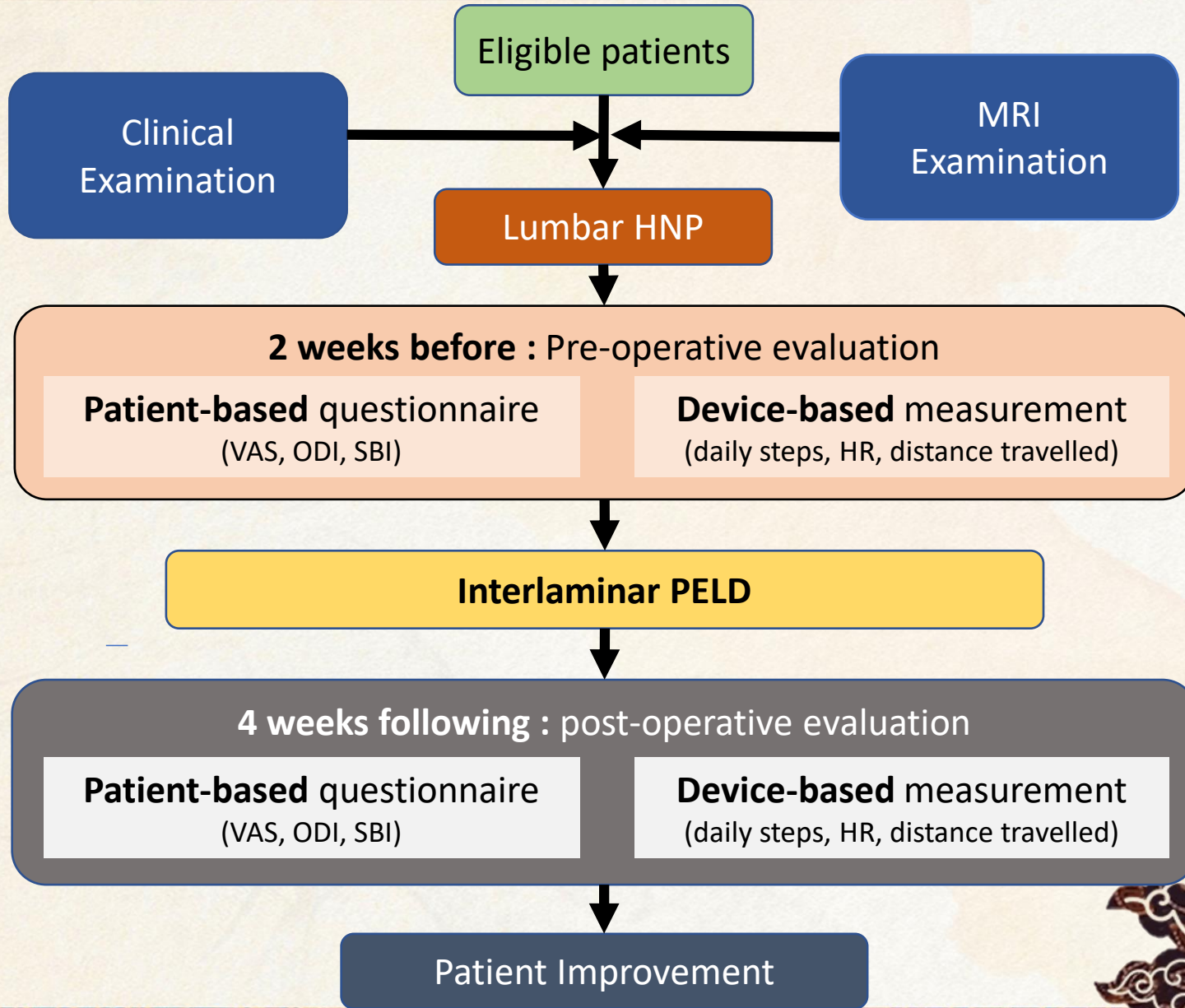
Inclusion Criteria

- Adult (>18 years old)
- Backpain due to HNP in lumbar spine, confirmed from clinical and MRI findings
- Undergone Index Interlaminar PELD procedure
- Participate in all evaluation prior and following the procedure

Exclusion Criteria

- Associated Fracture
- Backpain due to metastatic bone disease
- Spondylolisthesis
- Spinal stenosis of other causes
- Black disc adjacent to the herniated disc
- Previous lumbar surgery
- Assistive ambulatory device
- Arrhythmia and Cardiovascular disease
- Previously non ambulatory patient
- Special population (athlete, army)

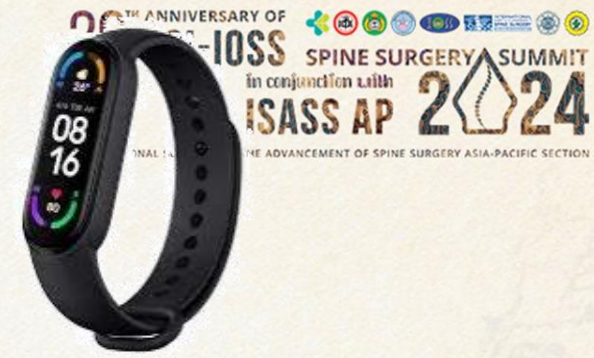
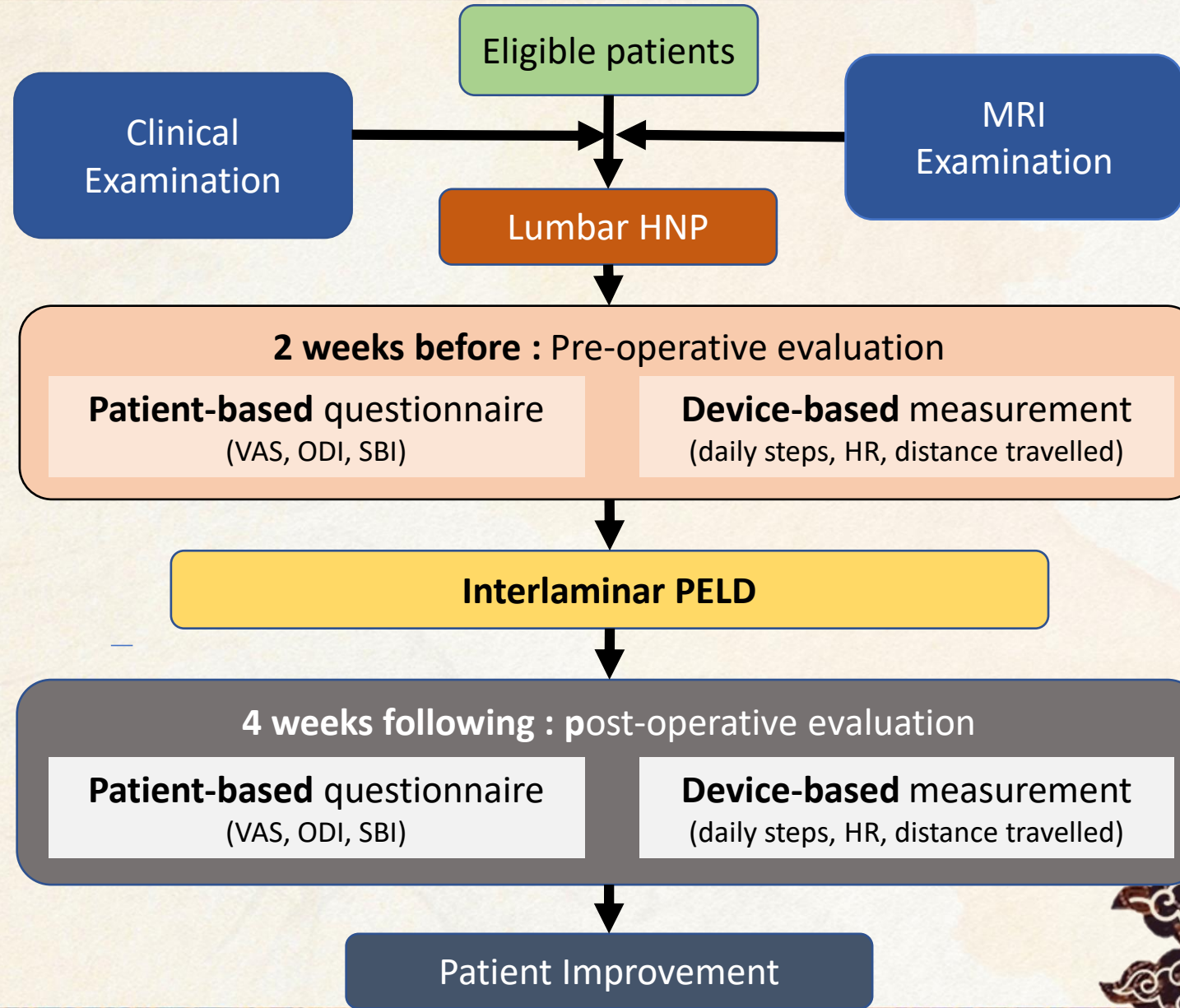
Research Process



Smart band used in this research



Research Process



Smart band used in this research

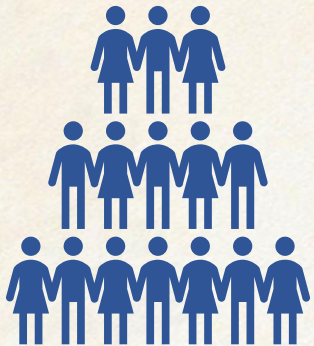
Analysis :

Correlation → **established (subjective) parameters** versus **wearable-based (objective) parameters**

Logistic regression → **significant parameters** to **predict** overall patient improvement

Results :

baseline characteristics



21
Patients

Demographic data	Value
Mean Age (years)	35.52 (±10.26)
Mean BMI (kg/m ²)	26.12 (±4.38)
Gender	
Male	11
Female	10
Smoking	11
History of injection therapy	7

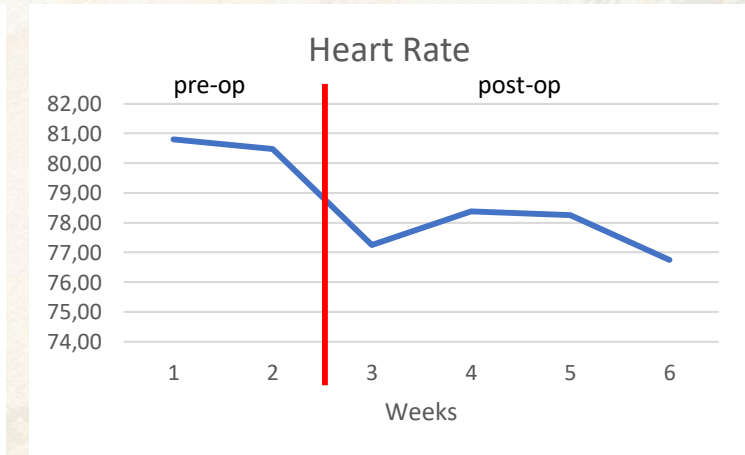
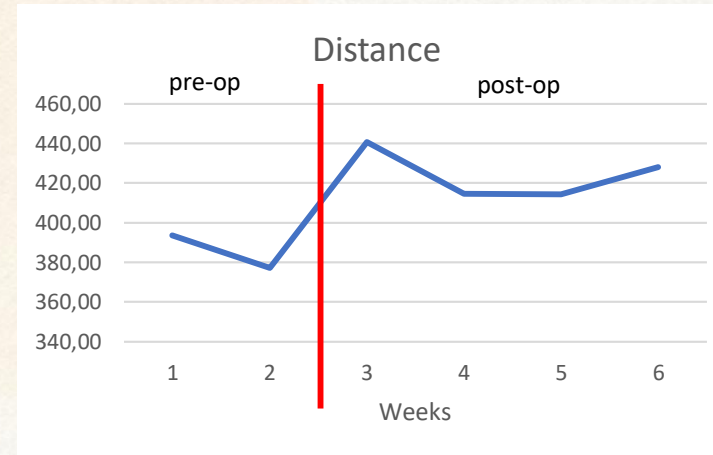
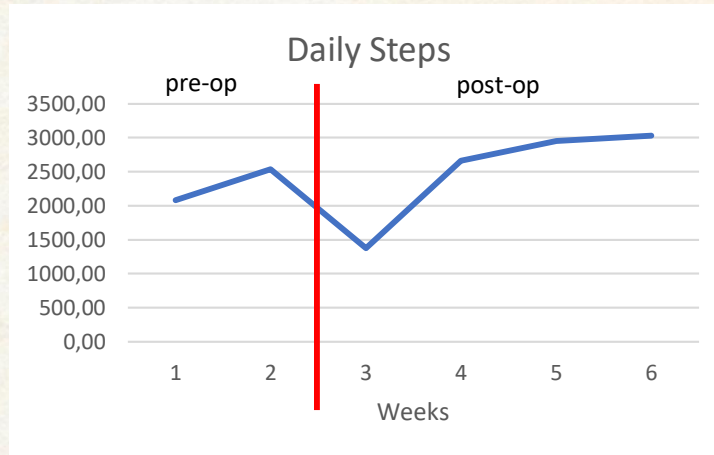
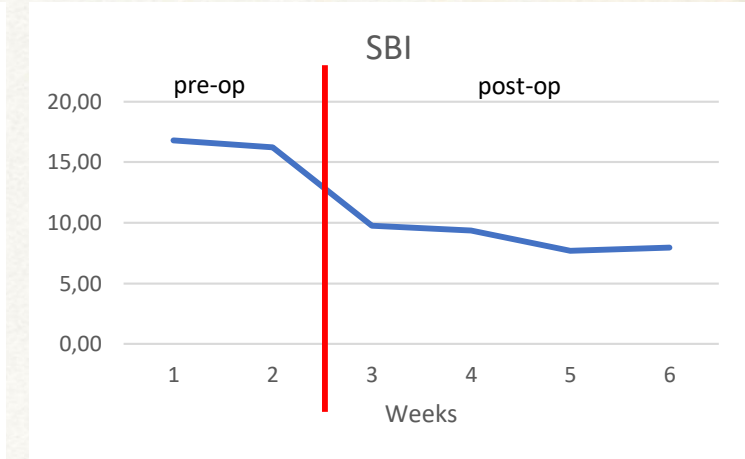
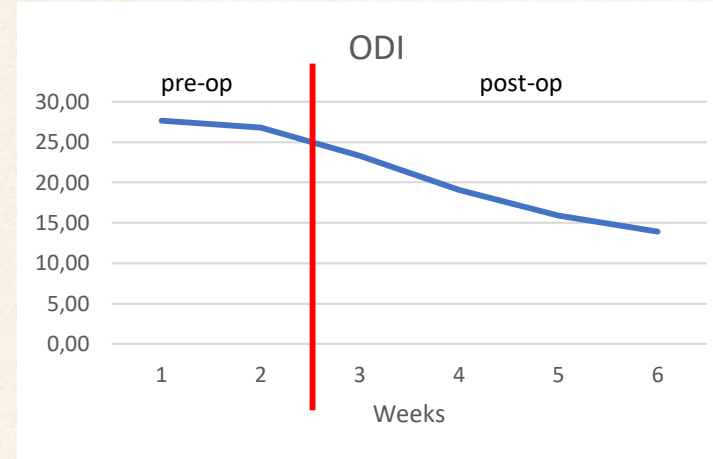
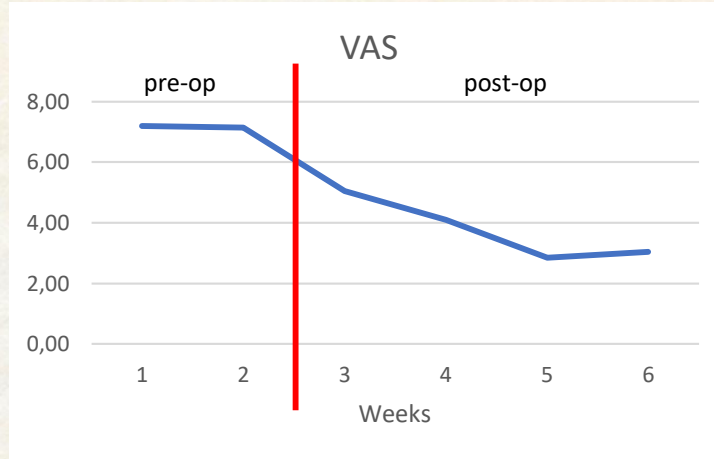
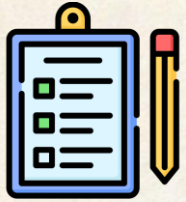
The measurement of both questionnaire-based and device-based parameters were **statistically significant** pre- and post-operatively

Parameters	Pre-Op	Post-Op	Changes	p-value
Established parameters :				
VAS	7.14 (±1.82)	3.05 (±2.01)	4.14 (±2.24)	.000*
ODI	27.62 (±10.22)	13.90 (±6.9)	13.71 (±7.9)	.002*
SBI	16.81 (±5.72)	7.95 (±5.75)	9.43 (±5.52)	.000*
Wearable device parameters :				
Daily steps	1906.52 (±1182.29)	4035.62 (±2013.49)	2189.45 (±1798.03)	.038*
Distance travelled (km)	0.85 (±0.53)	1.79 (±0.90)	0.94 (±0.79)	.026*
HR (bpm)	80.8 (±7.97)	76.74 (±7.85)	4.06 (±4.57)	.001*

*P<0.05

Results :

comparison of parameters pre- and post-op



Week 1-2 = pre-op measurement

Week 3-6 = post-op measurement



Results : baseline characteristics

Baseline Characteristics	p-value					
	VAS	ODI	SBI	Steps	Distance	HR
Age	.393	.152	.561	.514	.448	.820
BMI	.107	.461	.149	.078	.066	.442
History of Smoking	.789	.194	.631	.591	.558	.972
Gender	.789	.194	.631	.591	.558	.972
History of Injection Therapy	.153	.184	.872	.613	.544	.597

There were no significant correlation of all measured parameters with the baseline characteristics



No Confounding variables were found



Results : correlation study

Wearable technology parameters	VAS		ODI		SBI	
	Correlation Coefficient (r)	P Value	Correlation Coefficient (r)	P Value	Correlation Coefficient (r)	P Value
Steps	-0.450	0.041	-0.515	0.017	-0.714	0.000
Distance Travelled	-0.519	0.016	-0.494	0.023	-0.649	0.000
Heart Rate	-0.310	0.171	-0.281	0.217	-0.476	0.029

- Changes in steps and distance following PELD correlated with changes in VAS, ODI and SBI scores.
- However, average HR was only correlated with SBI → HR may be influenced by many factors



Results : correlation study

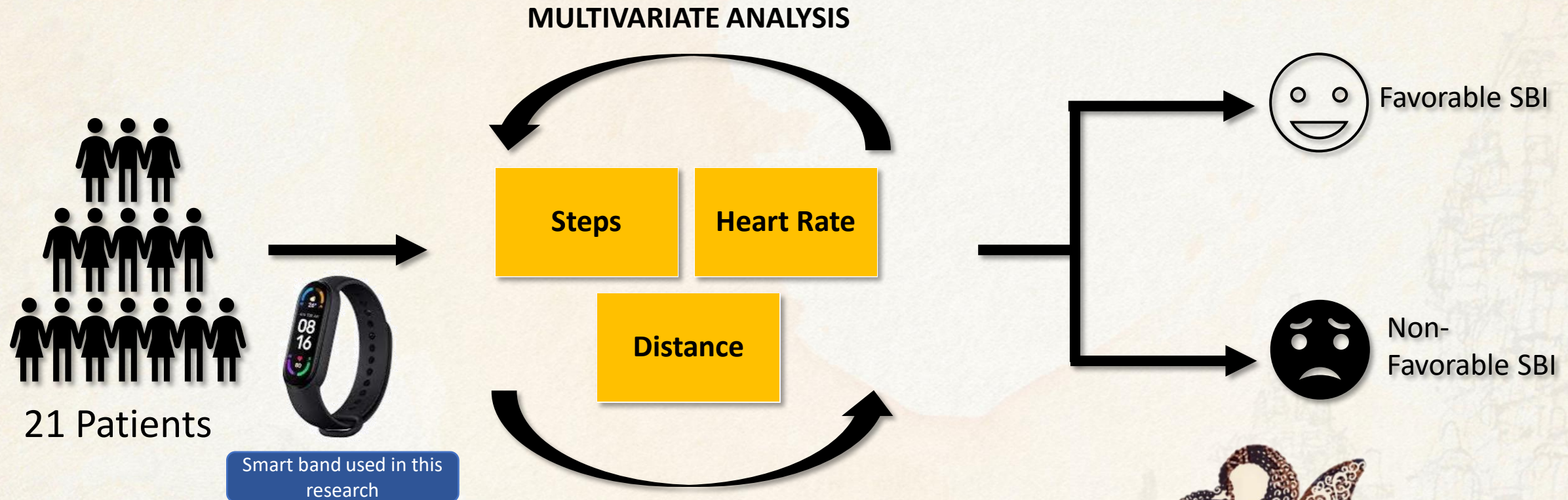
Wearable technology parameters	VAS		ODI		SBI	
	Correlation Coefficient (r)	P Value	Correlation Coefficient (r)	P Value	Correlation Coefficient (r)	P Value
Steps	-0.450	0.041	-0.515	0.017	-0.714	0.000
Distance Travelled	-0.519	0.016	-0.494	0.023	-0.649	0.000
Heart Rate	-0.310	0.171	-0.281	0.217	-0.476	0.029

- Changes in steps and distance following PELD correlated with changes in VAS, ODI and SBI scores.
- However, average HR was only correlated with SBI → HR may be influenced by many factors

Results of wearable technology parameters against SBI fulfill criteria to continue into multivariate analysis



Concept : linear regression study



Can wearable technology parameters predict the outcome of SBI to produce a scoring system?



Results : linear regression study

Wearable technology parameters	Linear Regression	
	Coeffisient	P Value
Heart Rate	-0.364	0.060
Daily Steps	-0.306	0.789
Distance Travelled	-0.227	0.843

Multivariate analysis showed **no significance** between change in HR, steps, and distance travelled following PELD with changes in SBI score

➔ Data is not supportive for production of composite scoring system



Discussion



Novelty

This is the **first study** to conduct **regression analysis** on wearable device-based parameters to predict postoperative patient **outcome and satisfaction** following **PELD surgery**

Previous studies have noted **significant relationship** between steps, distance travelled, and heart rate variability on postoperative patient outcome
→ indicating its **potentials**



Discussion

Although significantly correlated, multivariate regression analysis showed **failure to produce scoring system** based on parameters used in our study



May be due to **limitation of number of samples and follow up**

Mobbs et al. (2019) have previously proposed a scoring system based on step count, gait velocity, step length, posture → significant positive correlation with ODI, negative correlation with PSI (3 months follow up and 13 patients)

Stienen et al. (2020), reported increased activity following spine surgery was not statistically significant compared to pre-operative findings

Mobbs, R.J., Mobbs, R.R. and Choy, W.J., 2019. Proposed objective scoring algorithm for assessment and intervention recovery following surgery for lumbar spinal stenosis based on relevant gait metrics from wearable devices: the Gait Posture index (GPI). Journal of Spine Surgery, 5(3), p.300.

Stienen, M.N., Rezaii, P.G., Ho, A.L., Veeravagu, A., Zygourakis, C.C., Tomkins-Lane, C., Park, J., Ratliff, J.K. and Desai, A.M., 2020. Objective activity tracking in spine surgery: a prospective feasibility study with a low-cost consumer grade wearable accelerometer. Scientific Reports, 10(1), p.4939.



Discussion : why wearable technology?

Wearable technology is **increasingly recognized** for its potential to enhance and monitor post-operative patient outcomes

Enhanced Monitoring and Recovery

Wearable devices allow real-time tracking of patient activity → crucial to track early mobilization post-surgery

Objective vs Subjective Measures

Wearables allow for objective measurement of patient's activity compared to questionnaire-based parameters

Patient Perceptions and Acceptance

Patients showed a strong willingness and interest to use wearable technology → compliance and ease of use

Predictive Ability for Intervention

Integration of measurements allow higher judgement value of patient's condition → predictive for early intervention

While wearables provide objective data, there is a valuable insight obtained in using the questionnaire-based measurements → **combined approach is more effective to predict the outcome**

Conclusions :

- **Changes in steps and distance travelled** following PELD surgery was **significantly correlated** with changes in **subjective scoring** systems (ODI, SBI, VAS)
- Smart wearable technology is a **valuable tool** to be used **in conjunction** with **subjective parameters**
- Further studies required in identifying parameters to be used **to produce a scoring system** based on smart wearable technology parameters



References :

- Haugen, A.J., Grøvle, L., Brox, J.I., Natvig, B., Keller, A., Soldal, D. and Grotle, M., 2011. Estimates of success in patients with sciatica due to lumbar disc herniation depend upon outcome measure. *European Spine Journal*, 20, pp.1669-1675.
- Monticone, M., Baiardi, P., Vanti, C., Ferrari, S., Pillastrini, P., Mugnai, R. and Foti, C., 2012. Responsiveness of the Oswestry Disability Index and the Roland Morris Disability Questionnaire in Italian subjects with sub-acute and chronic low back pain. *European spine journal*, 21, pp.122-129.
- Sánchez-Margallo, J.A., Rabazo, J.C., de Miguel, C.P., Gloor, P., Rey, D.D., González-Portillo, M.R., Agudelo, I.L. and Sánchez-Margallo, F.M., 2021. Wearable Technology for Assessment and Surgical Assistance in Minimally Invasive Surgery. In *Advances in Minimally Invasive Surgery*. IntechOpen.
- Knight, S.R., Ng, N., Tsanas, A., Mclean, K., Pagliari, C. and Harrison, E.M., 2021. Mobile devices and wearable technology for measuring patient outcomes after surgery: a systematic review. *NPJ digital medicine*, 4(1), p.157.
- Mobbs, R.J., Mobbs, R.R. and Choy, W.J., 2019. Proposed objective scoring algorithm for assessment and intervention recovery following surgery for lumbar spinal stenosis based on relevant gait metrics from wearable devices: the Gait Posture index (GPi). *Journal of Spine Surgery*, 5(3), p.300.
- Stienen, M.N., Rezaii, P.G., Ho, A.L., Veeravagu, A., Zygorakis, C.C., Tomkins-Lane, C., Park, J., Ratliff, J.K. and Desai, A.M., 2020. Objective activity tracking in spine surgery: a prospective feasibility study with a low-cost consumer grade wearable accelerometer. *Scientific Reports*, 10(1), p.4939.





Thank You for your attention

